FACULTY OF COMPUTING
UTM Johor Bahru

Project Part #3: Conceptual and Physical Design (10%)

Subject	:	Human-Computer Interaction Fundamentals (SECM2113)
Session	:	2025/2026 Semester 1
Title	:	Project Part #3: Conceptual and Physical Design
Duration	:	Approx. 2 weeks
Submission	:	This is a **TEAMWORK.** The team must update the site accordingly, by the 25th of December 2025

NOTE TO STUDENTS

This part builds directly on Part 2: Establishing Requirements. You have clarified your user groups, scenarios, tasks, and design requirements for a human-centred digital twin interface. Now you will transform these requirements into conceptual and physical designs.

Your focus remains on HCI: designing clear, usable, and meaningful interaction experiences for users engaging with digital twin dashboards, alerts, predictive insights, or monitoring tasks.

==========

Process 2: Developing Alternative Designs

After identifying needs and requirements, this stage focuses on conceptualising and visualising the system's interaction through:

1. **Storyboards** (to show user interaction in context)
2. **Alternative interface ideas** (Crazy 8 sketches)
3. **Wireframes** (initial structure of your digital twin interface)
4. **Interaction metaphors** (to guide intuitive user experience)

1. Conceptual Design - Storyboards

Storyboards help envision how your **personas** will use the proposed digital twin system in their **real scenarios**. For each of the 3 tasks from Part 2, create a storyboard showing:

- The persona
- The context (industrial environment, workstation, control room, training lab, etc.)
- The interaction sequence
- The task goal and outcome

For the digital twin theme, storyboards might show:

- An operator responding to a sudden temperature alert
- A supervisor analysing predictive insights to schedule maintenance
- A trainee learning machinery behaviour through simulation dashboards

Use hand-drawn sketches. Scan or photograph them clearly.

2. Physical Design (Alternative Ideas) – Crazy 8

Each group member must perform **Crazy 8**. individually:

- 8 ideas in 8 minutes
- Fast sketches exploring different ways to visualize data, alerts, or dashboard layouts

Examples relevant to digital twin systems:

- Alternative alert visualisation (colour, shape, motion)
- Different dashboard layouts emphasizing clarity and cognitive load reduction
- Novel interaction metaphors (e.g., machine health represented as a living organism)

Upload **all members' Crazy 8 sketches**. Then as a group:

- Vote on the strongest elements
- Identify the most promising layout directions
- Post the "voted" combined design ideas

3. Wireframes – Interaction Flow for the 3 Tasks

Create wireframes showing how users will move through the interface to complete your 3 key tasks. Wireframes should:

- Include functional interface components (buttons, icons, menus, filters)
- Reflect the cognitive and functional needs discovered in Part 2
- Show clear interaction flow (screen-to-screen movement)

For digital twins, wireframes might include:

- Live dashboard view with machine status indicators
- Alert details and recommended action panel
- Predictive insight visualization (graphs, trends)
- Screen to customize or filter data view

Your justification must show alignment with:

- Gestalt Principles (e.g., proximity, similarity for grouping data)
- Usability goals & UX goals
- Shneiderman's Golden Rules. (e.g., consistency, informative feedback)

4. Interaction Metaphors

Identify and justify **five (5) or more interaction metaphors** that enhance user understanding. Examples for digital twin contexts:

- Health metaphor – machine status shown like patient vital signs
- Traffic light metaphor – system alerts using clear colour logic (green/yellow/red)
- Control room metaphor – dashboard organized like a real industrial console
- Digital mirror metaphor – interface reflects real machine behaviour
- Weather metaphor – predictive risk shown as storm severity levels

Justify why these metaphors help your user groups interpret information more intuitively.

Submission 3: Conceptual & Physical Design

Report should include the following:

- **Storyboards**. For each of your **THREE (3) user tasks/functions**, demonstrate how your proposed app would be used by your persona within the scenario that you described in Project #2. **Hand-sketching is preferred. This means you will have to scan/take a picture of the storyboard and upload.** (8m)

- **Alternative Designs.** Each group member should produce sketches of their ideas for interface design through the Crazy 8 exercise. Post **each** members' Crazy 8 sketches. (6m)

- **Wireframes**. Post your wireframe design that could roughly show the flow of **each** three tasks. **Justify** your design decisions. Provide description on how your **design requirements** from Project #2 is incorporated in the designs. (12 m)

- **Interaction Metaphors**. Describe and justify the metaphors that your group deemed to be the most suitable interaction metaphors for your interface. List and justify **5 or more** type of metaphor. (6 m)

Reference – Topic 3 Metaphors, Design Principles, Topic 8 Prototyping & Construction

Further readings and videos on Crazy 8 method:
https://youtu.be/KgJCk5sRdoU
https://youtu.be/yz4g87XapQ0
https://youtu.be/UXOLJy0E7Pg
https://youtu.be/ccOuPN-Z8ls

https://www.iamnotmypixels.com/how-to-use-crazy-8s-to-generate-design-ideas/
https://blog.prototypr.io/how-to-run-a-crazy-eights-workshop-60d0a67b29a

Further readings on good examples of UI/UX metaphors and analogies:
https://uxdesign.cc/interaction-metaphors-91a723aea4e1
https://medium.com/strv-design/why-metaphors-matter-for-app-designers-2fb477854f66
https://blog.prototypr.io/some-useful-and-some-less-useful-icon-metaphors-for-ui-ad225e4fef0a

Scoring Rubric:

Item	Absent (0)	Minimal (1)	Moderate (2)	Good (3)	Excellent (4)	Weight	Score
Storyboard	No storyboard	Demonstrate only 1 goal through hand sketching. Sketches either task, user, context – but not all	Demonstrate only 2 goals through hand sketching. Sketches either task, user, context – but not all	Demonstrate 3 goals through hand sketching. Sketches either depicts task, user, context – but not all	Demonstrate 3 goals clearly through hand sketching. Sketches depicts task, user, context	2	/8
Alternative Designs	No Crazy 8 sketches	More than two member's Crazy 8 sketches missing	Two member's Crazy 8 sketches missing	One member's Crazy 8 sketches missing	All group member's Crazy 8 sketches were provided	1.5	/6
Layout & Interface Design (wireframe)	No UI design	Each interface design has no control that user can manipulate and interact with. Provide no justification	Each interface design has little control that user can manipulate and interact with. Provide minimal justification	Each interface design has clear but not so distinguished control user can manipulate and interact with. Provide moderate justification	Each interface design has clear and distinguished control user can manipulate and interact with. Provide strong justification using HCI principles	3	/12
Interaction Metaphors	No metaphors provided	Provided less than 5 interaction metaphors with/without justification	Provided 5 or more interaction metaphors with some questionable justification.	Provided 5 or more interaction metaphors with some good justification.	Provided 5 or more interaction metaphors, all with excellent justification.	1.5	/6

32/32 = 10%